

Mathematical Audit of the DRCPT-PG Draft

Independent proof audit and Lean verification

9 September 2026

Verdict

FAIL in the current form. The model is coherent, and several local arguments are valid under explicit regularity conditions. The projected-ascent lemma is false as stated. Consequently, the dynamic-local-regret theorem and its downstream corollaries are not established. The minimax proof also lacks a CPT-realizable lower-bound construction.

Framework

Episode k starts from state, wealth X , reference r , previous portfolio ω , and factor parameter θ . Bounded asset features are mapped to Dirichlet concentrations. The execution layer uses the policy mean; stochastic learning trajectories produce terminal relative wealth $Y_k = X_k - r_{k,h}$. The distribution of Y_k defines $J_k(\theta, r_k)$. A score identity represents ∇J_k as an expected CPT multiplier times a trajectory score. A proposed multilevel estimator drives a normalized projected update. Market and reference drift move the stationary set.

Result Audit

Claim	Status	Reason
Path dependence	Verified	Algebra checked in Lean.
CPT score-gradient identity	Conditional	Requires DQM, domination, and adequate score moments.
Plug-in bias $O(n^{-1})$	Incomplete	The Taylor remainder needs stronger smoothness.
Multilevel unbiasedness	Conditional	Telescoping is valid after the coupled statistic is fully defined.
Variance and CLT	Conditional	Need base-level moments and consistent conditioning.
Stationary-set drift	Valid with clarification	Requires a symmetric uniform gradient-drift statement.
Projected ascent	False	Boundary projection can remove a first-order ascent step.
Dynamic local regret	Not established	Relies on the false projected-ascent lemma.
Stationary-set tracking	Partial	Error-bound inequality is valid; raw-gradient control is missing.
Minimax CPT rate	Not established	Abstract product laws are not embedded in the CPT trajectory model.

Decisive Counterexamples

For $\Theta = [0, 1]$, $J(\theta) = \theta$, $\theta = 1$, $d = 1$, and $\gamma > 0$, projection gives $\Pi_{\Theta}(1 + \gamma) = 1$. Thus the objective increment is zero, while the claimed lower bound contains a positive first-order term γ and only a quadratic deduction.

For a two-period equal-weight smoothing window with gradients $g_1 = 1$ and $g_2 = -1$, the smoothed gradient is zero while $g_1^2 + g_2^2 = 2$. Dynamic local regret based on smoothed gradients therefore cannot by itself control every raw gradient or distance to every current stationary set.

Required Repairs

1. Use an unprojected bounded-domain analysis, or replace the update analysis with a projected-gradient-mapping theorem.
2. Replace the almost-sure bounded Dirichlet score by uniform moment assumptions, or change the policy support.
3. Fully define cross-fitting, shared scores, nested level coupling, and base-level moments.
4. Separate market drift, simulation mismatch, and within-episode drift.
5. Prove raw-gradient control separately from smoothed dynamic local regret.
6. Normalize the endpoint-regularized probability weights and specify the zero-region smoothing function.
7. Add a fixed-objective condition to the static corollary.
8. Construct a CPT-realizable minimax submodel or narrow the lower-bound claim.

Formal Verification

The companion Lean 4.33.1 project builds successfully with `lake build`. It contains no `sorry`, `admit`, or undeclared axioms. Its scope is finite-dimensional rational algebra and executable counterexamples; measure-theoretic and asymptotic claims remain subject to the paper audit.